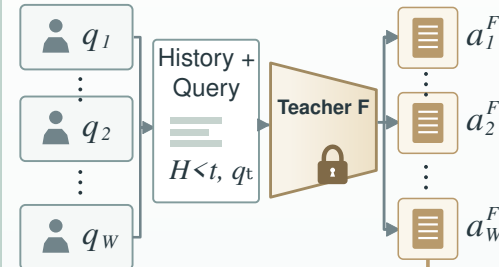
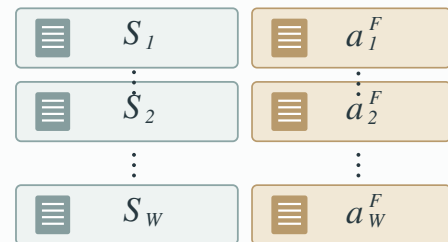


# 1 Warm-Start

Early dialogue turns  
(1 ... W)



## Warm-Start Pairs

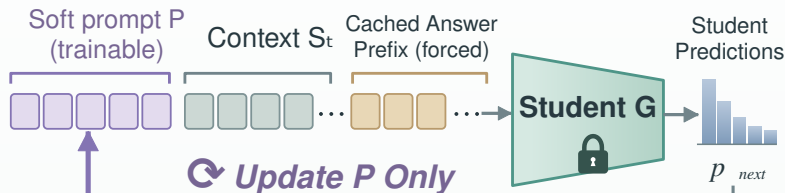


Cache Teacher Replies

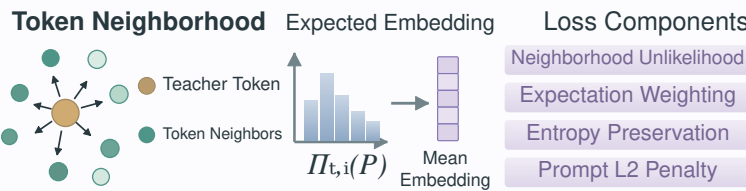
# 2 Soft-Prompt Mining

Discard  
After Mining

## Input assembly (teacher-forced)



Mining Objective (teacher-forced student predictions vs. cached targets)



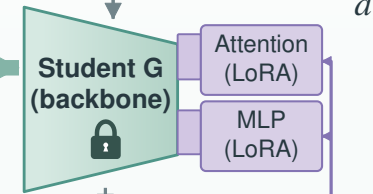
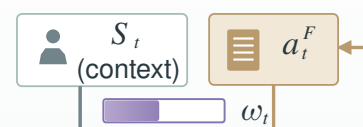
All Warm-Start Turns · Weights  $\omega_t \propto r_t$



Optimize P · Freeze G · Anchoring Scores  $r_t$

# 3 Local LoRA Distillation

All W warm-start turns  
(anchoring weights  $\omega_t \propto r_t$ )



$$\mathcal{L}_{FT} = 1/W \sum_t \omega_t NLL_t$$

$$NLL_t = -\log G_{\theta}(a_t^F | S_t)$$

Update LoRA Only

Adapted Student  $G_{\theta}$

One-Time Adaptation

# 4 Gated Serving

## Acceptance Checks

- Response Fidelity (acceptance batch vs. F)
- Query Locality (vs. session centroid)

Both Pass?

Yes

No

Recheck

Teacher F Full history

Rollback

$m_{cons}$  consecutive violations

Refresh Summary + Centroid

Updated context

Prompt-Free Serving · No Online LoRA Updates

Summary + Recent K Turns + Query

...

Reply

Recent queries

Drift Monitor

Sliding window

Updated context

Session timeline



Frozen



Offline update



Rollback